Kostiantyn Pysanyi

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Professional Summary

Computer Vision & Machine Learning Engineer with 3+ years delivering production medical-imaging systems and a background in civil engineering. Experience spans 2D–3D registration, mesh processing, and cloud hosting of image classification models; core tools: **Python** and **PyTorch**.

Experience

Machine Learning & Computer Vision Engineer Jun 2023 – Jul 2026
RAYLYTIC GmbH – Medical imaging & clinical-trial software Full-time

- Built an end-to-end **2D–3D registration** pipeline for implant pose estimation from X-rays, including **mesh processing and alignment**, used in clinical studies with typically 2,000–3,000 images per study; improved acceptance rate by 45% and reduced error by 15%; deployed with GPU-accelerated **Docker** and **Prefect**
- Implemented an **MLOps pipeline** combining data versioning with DVC, **MLflow** experiment tracking, and automated evaluation
- Delivered production **detection, segmentation, and classification** models for X-ray imaging; improved evaluation metrics (mAP, Dice, ROC) by 20–50% over baseline and reduced training time 8-fold
- Trained a DINOv3 image encoder self-supervised on X-ray data, then used it in a customized ViTDet architecture for supervised instance detection training on labeled data
- Migrated segmentation model serving to a centralized **Triton Inference Server** REST API, cutting latency by 50%
- Developed **3D instance segmentation and registration** for light-sheet microscopy images in a **liver cancer research project**.

Freelance Computer Vision Engineer Feb 2026 – Mar 2026
Smoothr.de / Sterlix GmbH Contract

- Developed **depth-map–based** camera obstruction detection for real-time retail anti-theft systems

Research Engineer Jan 2022 – Mar 2023
Data Science Research Group Part-time

- Designed data pipelines and training workflows for a ticket-routing classifier, transforming unstructured text into structured features and streamlining dataset updates and model evaluation for **industry partners**.
- Developed a demo ML model for bio-signal time-series classification using a high-performance computing cluster for pain-level assessment in surgical scenarios with non-responsive patients

Internships

- ML Thesis Student at RAYLYTIC GmbH – (Oct 2022 – Feb 2023): Developed anatomical landmark localization model
- ML Intern at Prudsys GmbH – (Dec 2021 – Feb 2022): Automated analytic workflows and data preprocessing pipelines
- Autonomous Driving Intern at IAV GmbH – (Sep 2021 – Nov 2021): Investigated inverse reinforcement learning for self-driving
- Student Research Assistant at WHZ – (Jan 2020 – Dec 2021): Developed a **semantic segmentation model for automated tissue-density analysis of 3D lung CT scans**; developed an autoML library and predicted patient admission to the ICU during COVID.

Technical Skills

Programming Languages: **Python**, C++17, TypeScript, Java, MATLAB

ML & CV: **PyTorch**, PyTorch Lightning, MONAI, TensorFlow, JAX, Detectron2, MMEngine/MMCV/MMDetection, Albumentations, scikit-learn, XGBoost, OpenCV, ITK, pyclesperanto, NumPy, pandas, CVAT, Hugging Face Transformers, timm

Experimentation & Tracking: **MLflow**, DVC, Jupyter Notebook/JupyterLab, TensorBoard, Optuna, Matplotlib/seaborn/Plotly

Infrastructure & Deployment: **Docker**, **Kubernetes**, **conda**, mamba, Prefect, AWS, Linux, CI/CD (GitLab, Bitbucket), Triton Inference Server, TensorRT

Backend & Dev Tools: **Git**, FastAPI, Flask, REST, OpenAPI, FHIR, SQL, MQTT, Codex, Claude Code

Education

B.Sc. Data Science with distinction (1.1 / 1.0) Oct 2019 – Apr 2023
University of Applied Sciences Zwickau Zwickau, Germany

B.Sc. and M.Sc. Civil Engineering (M.Sc. wGPA: 90.12/100) Sep 2014 – Jun 2020
Odesa State Academy of Civil Engineering and Architecture Odesa, Ukraine

Publications & Awards

EFORT 2026: Validation of a 2D-3D Image Registration Method for TKA Implant Pose Estimation. Pysanyi et al.

DWG 2023: Automatic determination of lumbar segmental lordosis based on an AI algorithm. Pysanyi et al.

1st Place: National Strength of Materials Olympiad (Ukraine, 2018)

Languages

German (C2; TestDaF C1, 2019), English (C2, EF SET Certificate)